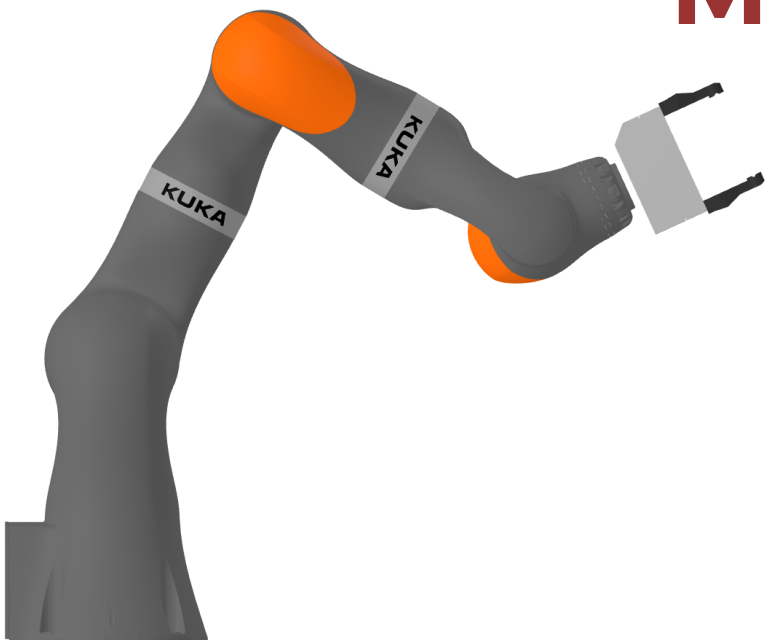


Motion Planning

Part I: Local Optimization

MIT 6.4210/2

Robotic Manipulation

Fall 2025, Lecture 10

Massachusetts Institute of Technology

Failure modes of "Clutter Clearing" demo

We can group them into:

1. Motion planning. Collisions and DiffIK limitations.
2. Perception.
    - grasp far from object center of mass,
    - double-picks,
    - phantoms...
3. More than pick and place.
    - objects stuck in the corner
    - objects too big for the hand

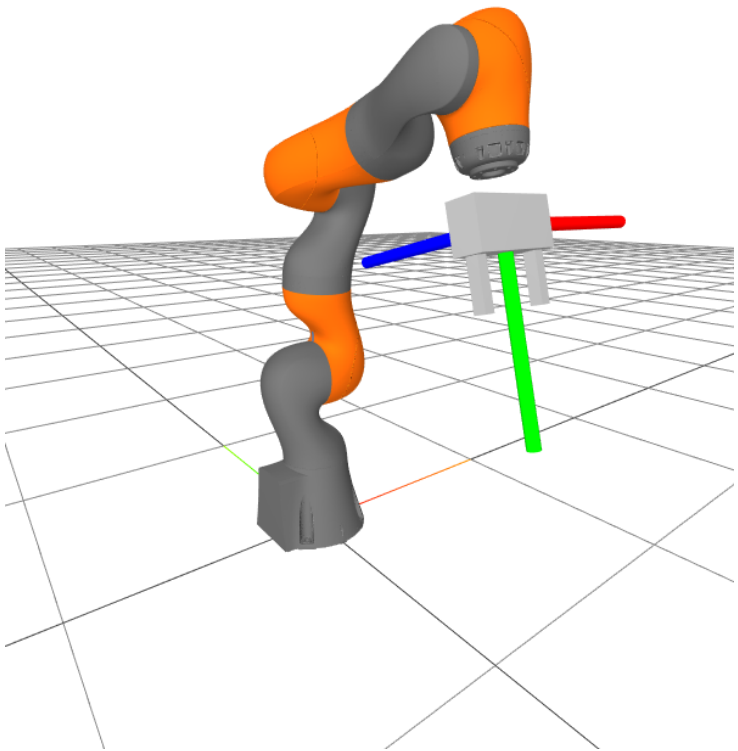
Motivation: Moving fast! (at the dynamic limits)





"Using trajectory optimization made scooping up to twice as fast"

credit: DEXAI ROBOTICS



$$X^B = f_{kin}^B(q)$$

Inverse kinematics (a rich history!)

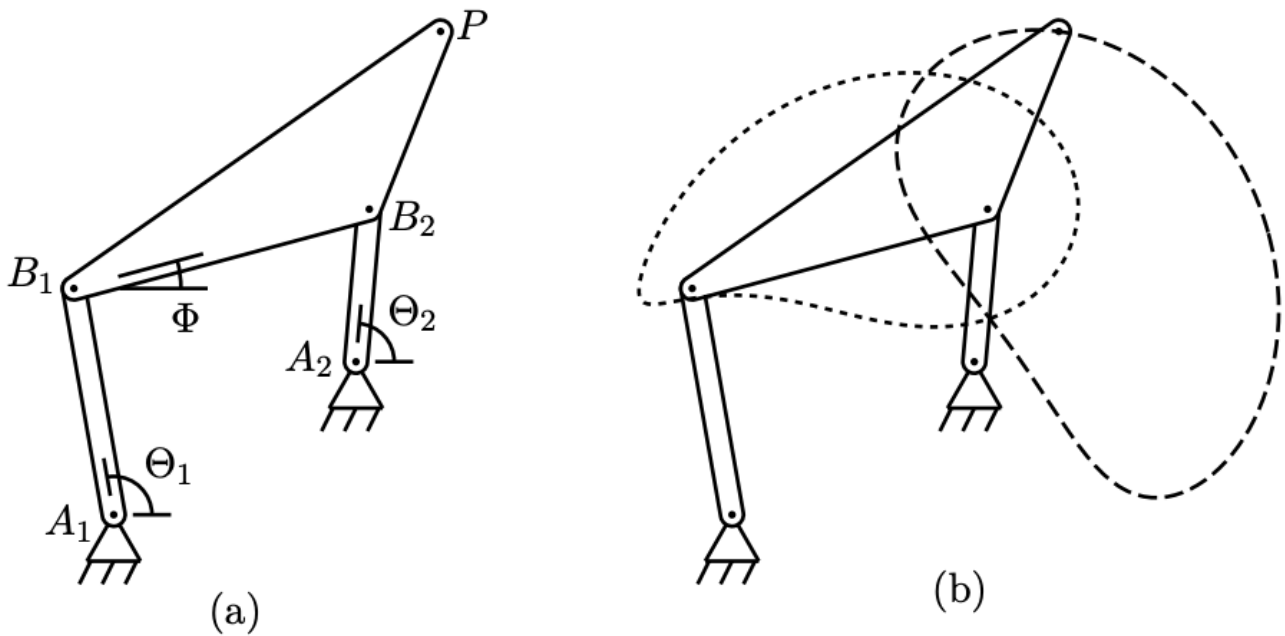
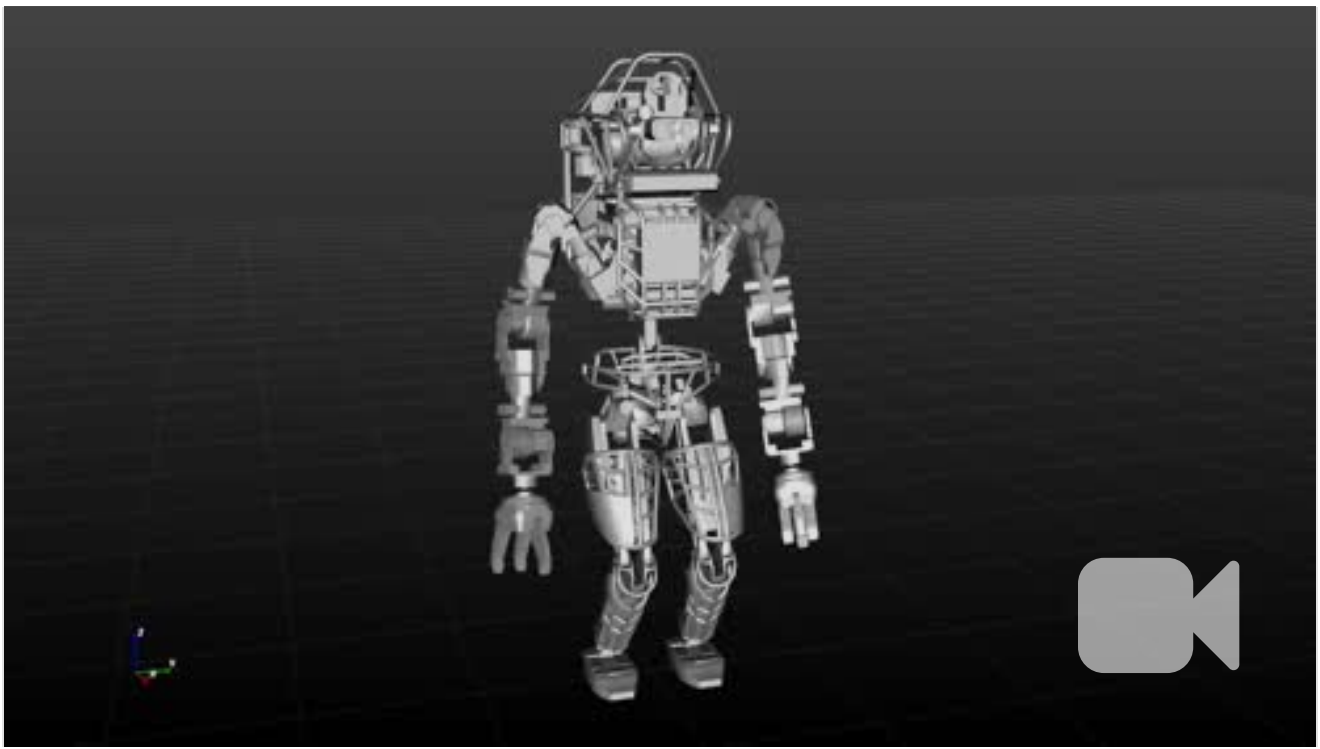


Figure 4: Four-bar linkage.

Charles W. Wampler and Andrew J. Sommese. Numerical algebraic geometry and algebraic kinematics. Acta Numerica, 20:469-567, 2011.

Inverse kinematics as an optimization



$$\min_q \left| q - q_{desired} \right|$$

subject to:

- rich end-effector constraints
- joint limits
- collision avoidance
- "gaze constraints"
- "feet stay put"
- balance (center of mass)
- ...

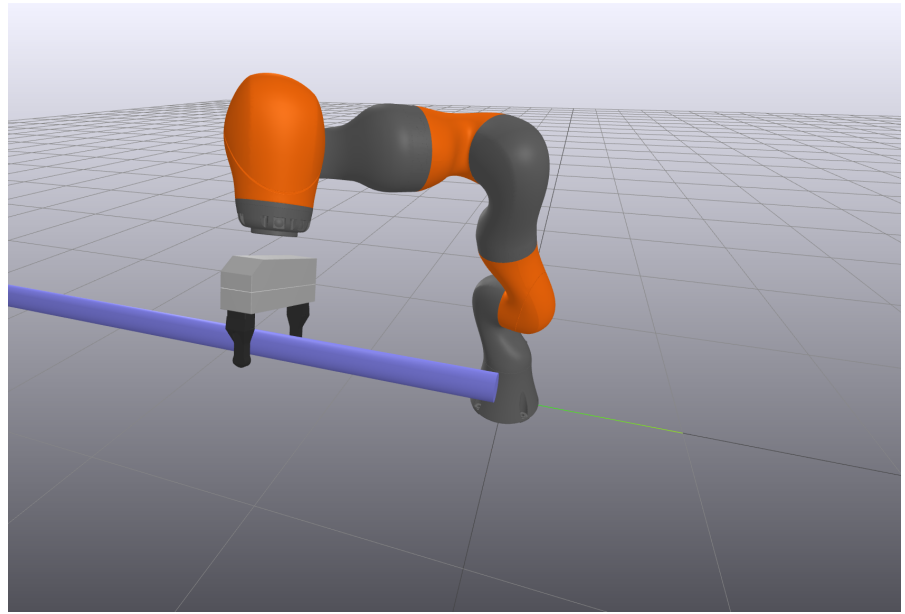
https://drake.mit.edu/doxygen_cxx/classdrake_1_1multibody_1_1_inverse_kinematics.html

```
        ik = InverseKinematics(plant, plant_context)
        ik.AddPositionConstraint(
            gripper_frame,
            [0, 0, 0],
            plant.world_frame(),
            pose.translation(),
            pose.translation(),
        )
        ik.AddOrientationConstraint(
            gripper_frame,
            RotationMatrix(),
            plant.world_frame(),
            pose.rotation(),
            0.0,
        )
        prog = ik.get_mutable_prog()
        q = ik.q()
        prog.AddQuadraticErrorCost(np.identity(len(q)), q0, q)
        prog.SetInitialGuess(q, q0)
        result = Solve(ik.prog())
```

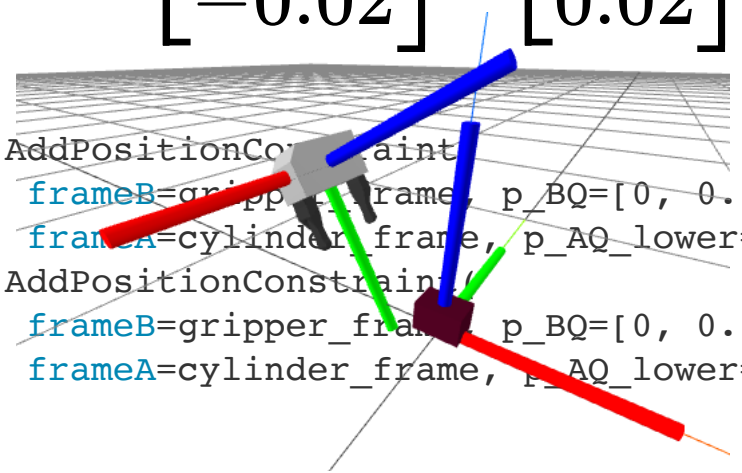
https://drake.mit.edu/doxygen_cxx/group__solver__evaluators.html

$$\begin{bmatrix} 0 \\ 0 \\ -0.5 \end{bmatrix} \leq {}^{C}X^{G}\,{}^{G}p^{Q} \leq \begin{bmatrix} 0 \\ 0 \\ 0.5 \end{bmatrix}$$

where C - cylinder, G - Gripper



for ${}^{G}p^{Q} = \begin{bmatrix} 0 \\ 0.1 \\ -0.02 \end{bmatrix}, \begin{bmatrix} 0 \\ 0.1 \\ 0.02 \end{bmatrix}$



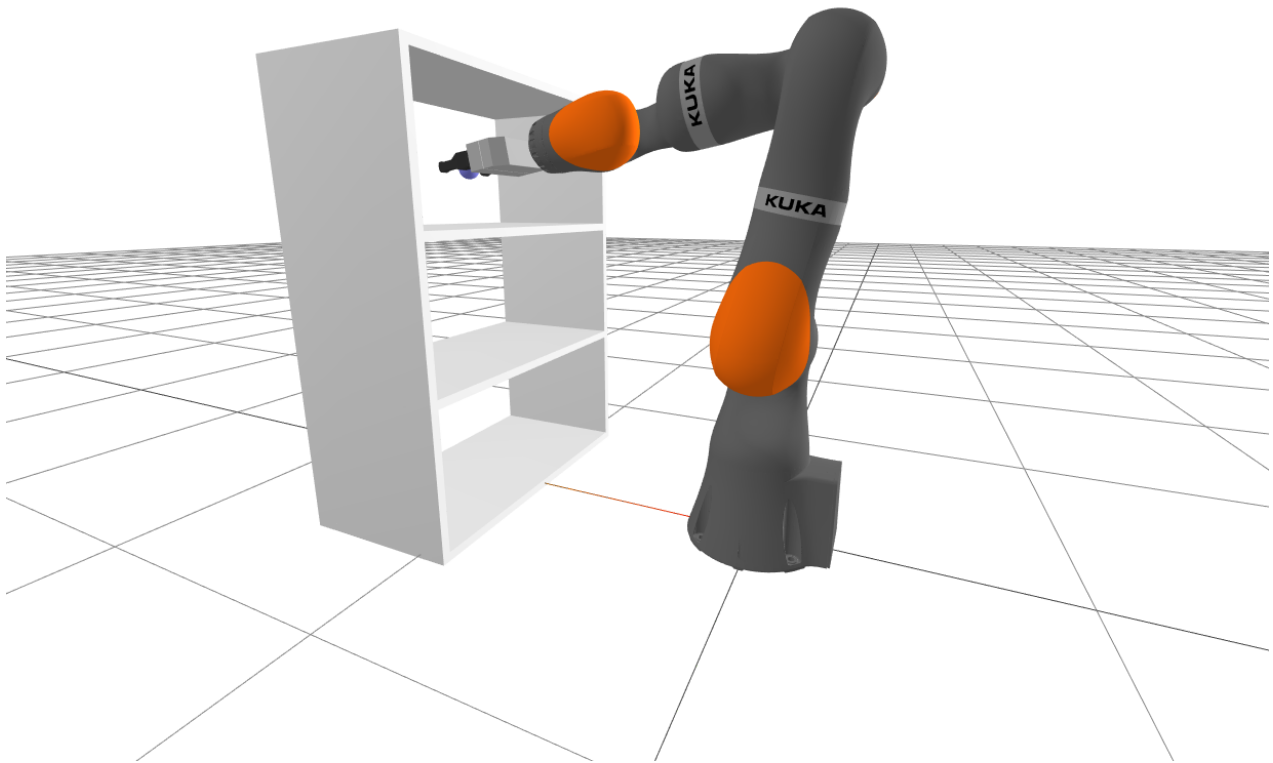
```
1 ik.AddPositionCo    aint
2     frameB=gripp    rame, p_BQ=[0, 0.1, -0.02],
3     fra A=cylinder_fra e, p_AQ_lower=[0, 0, -0.5], p_AQ_upper=[0, 0, 0.5])
4 ik.AddPositionConstrain
5     frameB=gripper_fra    p_BQ=[0, 0.1, 0.02],
6     frameA=cylinder_frame,  AQ_lower=[0, 0, -0.5], p_AQ_upper=[0, 0, 0.5])
```

https://drake.mit.edu/doxygen_cxx/classdrake_1_1multibody_1_1_minimum_distance_lower_bound_constraint.html

https://www.youtube.com/embed/m1rv4d_zUCY?enablejsapi=1&mute=1&rel=0

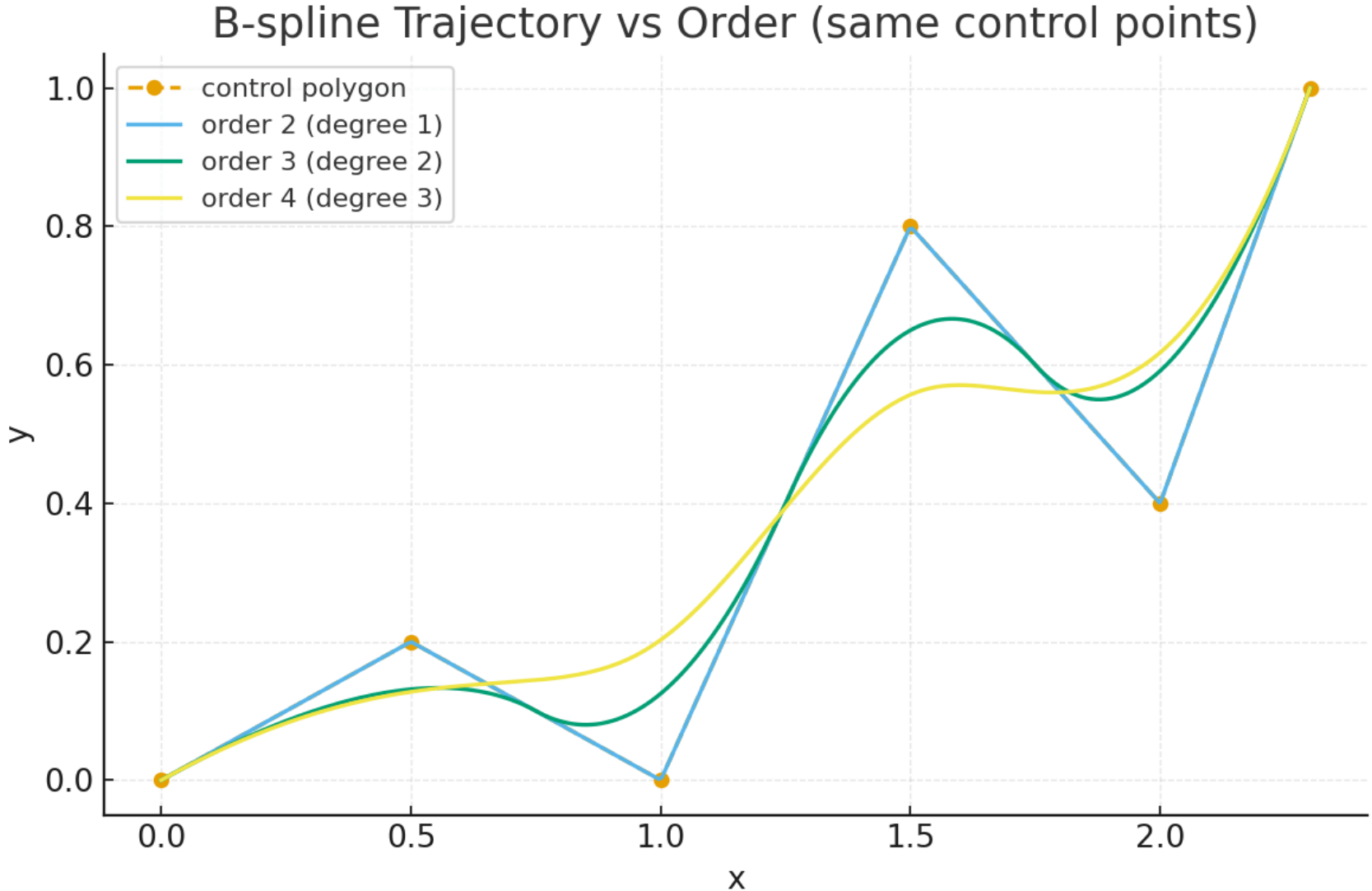
https://www.youtube.com/embed/E_CVq0lWfSc?enablejsapi=1&mute=1&rel=0

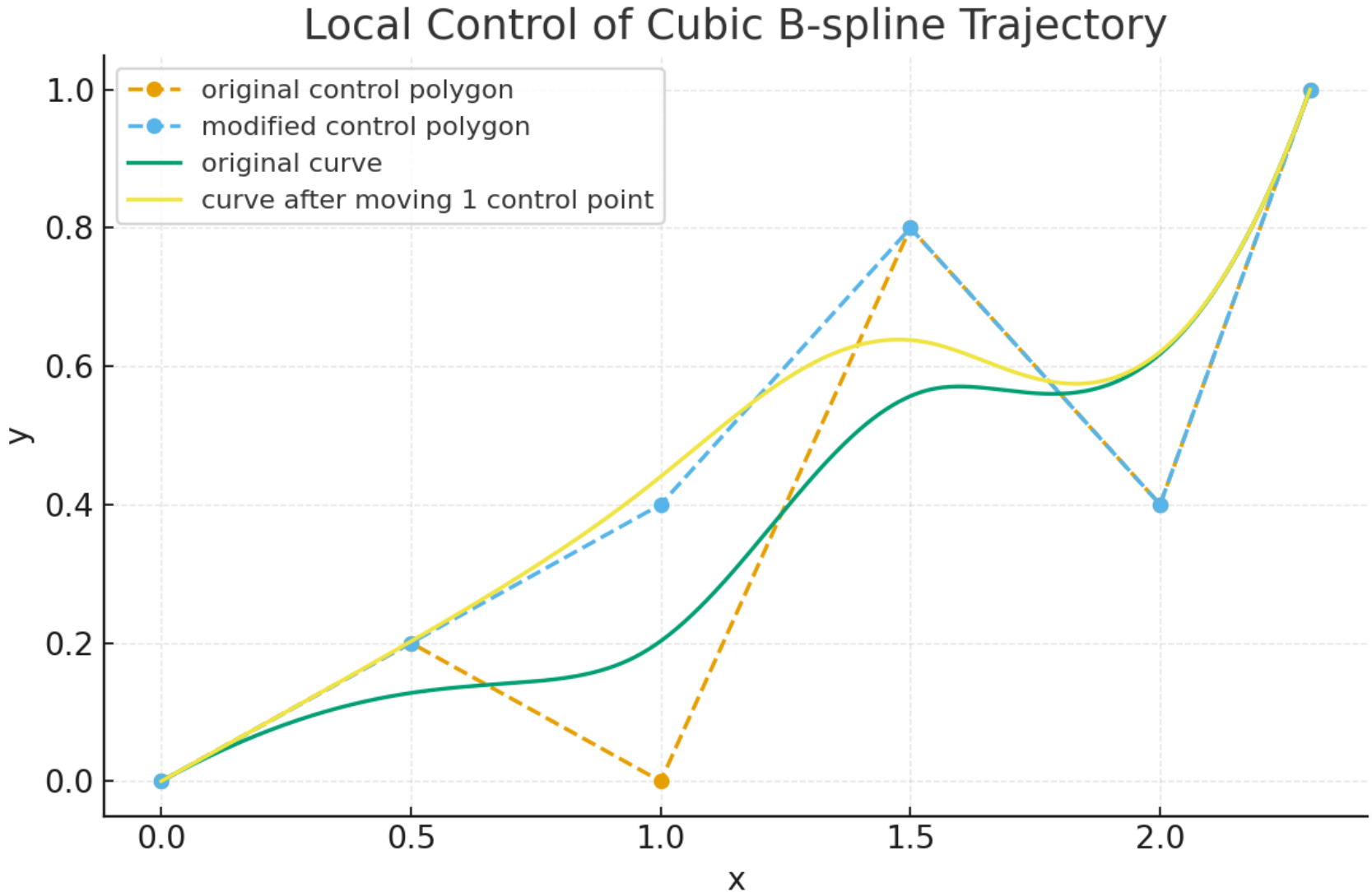
https://www.desmos.com/calculator/kol7mvpgke


KUKA
KUKA

https://manipulation.csail.mit.edu/data/iiwa_shelves_configuration_space.html

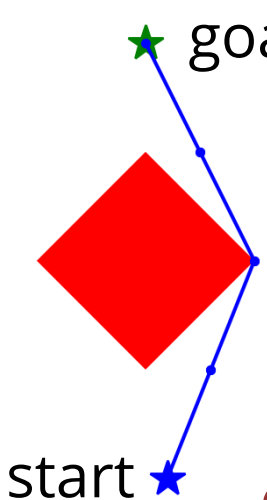
https://manipulation.csail.mit.edu/data/shelf_ik_prog.html


B-spline Trajectory vs Order (same control points)
control polygon
order 2 (degree 1)
order 3 (degree 2)
order 4 (degree 3)
1.0
0.8
0.6
0.4
0.2
0.0
y
0.0
0.5
1.0
1.5
2.0
x


Local Control of Cubic B-spline Trajectory
original control polygon
modified control polygon
original curve
curve after moving 1 control point
1.0
0.8
0.6
0.4
0.2
0.0
y
0.0
0.5
1.0
1.5
2.0
x

https://manipulation.csail.mit.edu/data/kinematic_trajectory_optimization_karate.html

Motion planning as a (nonconvex) optimization



fixed number of samples

$$\min_{q_0,\ldots,q_N} \sum_{n=0}^{N-1} |q_{n+1} - q_n|_2^2$$

$$\text{subject to} \quad q_0 = q_{start}$$

$$q_N = q_{goal}$$

collision-avoidance → $|q_n|_1 \geq 1 \quad \forall n$

(outside the L^1 ball)

nonconvex